

**Foundation Mathematics 1017SCG**  
**Week 9 Workshop Questions**

1. State the amplitude, period (in degrees), minimum value of  $y$  and maximum values of  $y$  for each of the following functions. Sketch each of the following functions, showing at least one full period (starting at  $x = 0^\circ$ ).

(a)  $y = 5 \sin(x)$

(b)  $y = 6 \cos(2x)$

(c)  $y = 3 \cos(4x) - 2$

(d)  $y = -4 \sin(2x) + 1$

(e)  $y = 2 \sin\left(\frac{x}{2}\right) + 1$

(f)  $y = 3 \cos\left(\frac{x}{4}\right) - 3$

2. Sketch each of the following functions (in radians) showing at least one full period (starting at  $x = 0$ ).

(a)  $y = 2 \sin\left(x - \frac{\pi}{2}\right)$

(b)  $y = 3 \cos[2(x + \pi)]$

(c)  $y = -\cos\left[\frac{1}{2}\left(x + \frac{\pi}{2}\right)\right]$

(d)  $y = \sin(4x - \pi)$

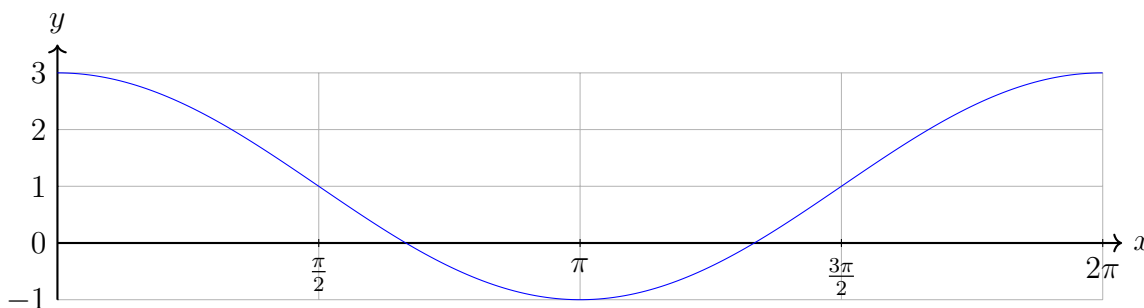
*Hint for part (d): The horizontal shift is NOT  $\pi$ . To determine horizontal shift, the equation must first be in the form  $y = A \sin[B(x + C)] + D$ .*

3. Sketch each of the following functions.

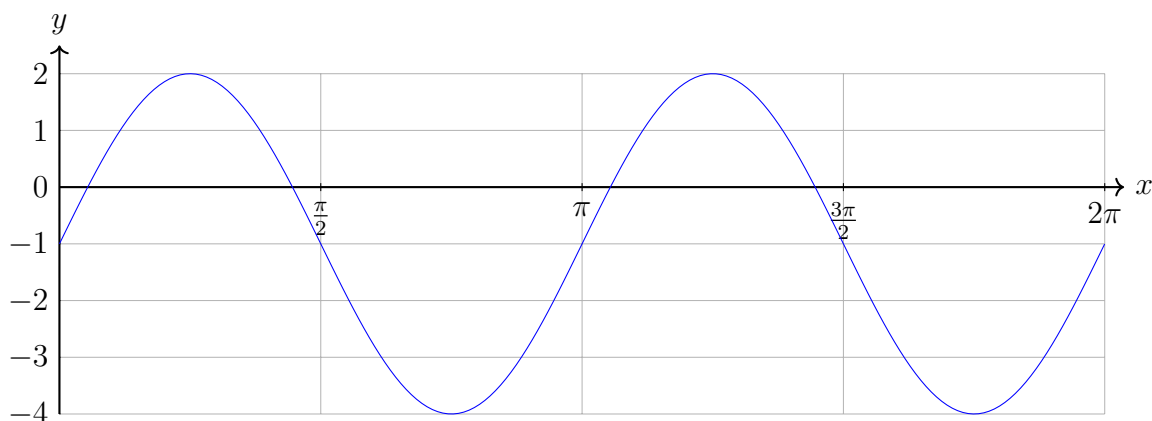
(a)  $y = 3 \sin\left[2\left(x - \frac{\pi}{2}\right)\right] + 4$  for  $0 \leq x \leq 2\pi$

(b)  $y = -6 \cos\left(x + \frac{\pi}{2}\right) - 2$  for  $-2\pi \leq x \leq 2\pi$

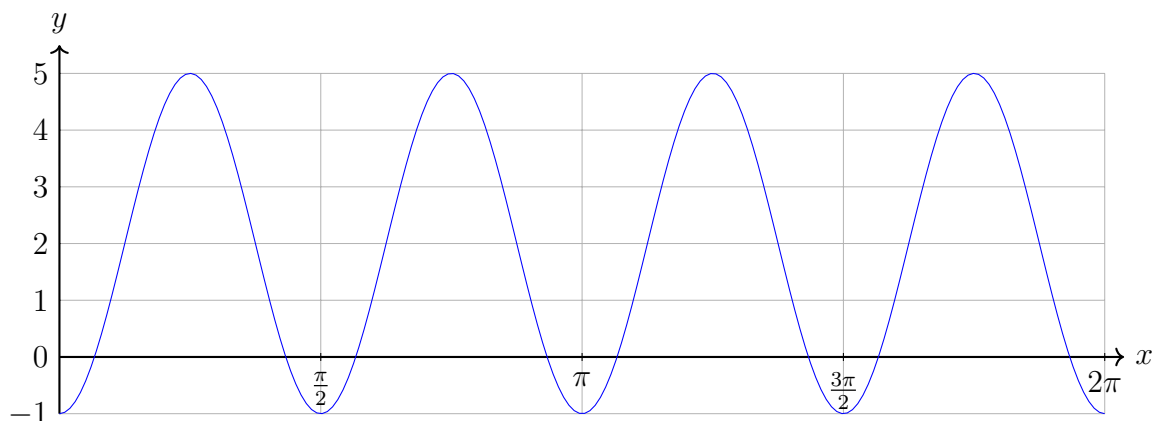
4. (a) Find an equation for the graph below in the form  $y = A \sin(Bx) + D$  or  $y = A \cos(Bx) + D$  (where  $A$ ,  $B$  and  $D$  are constants).



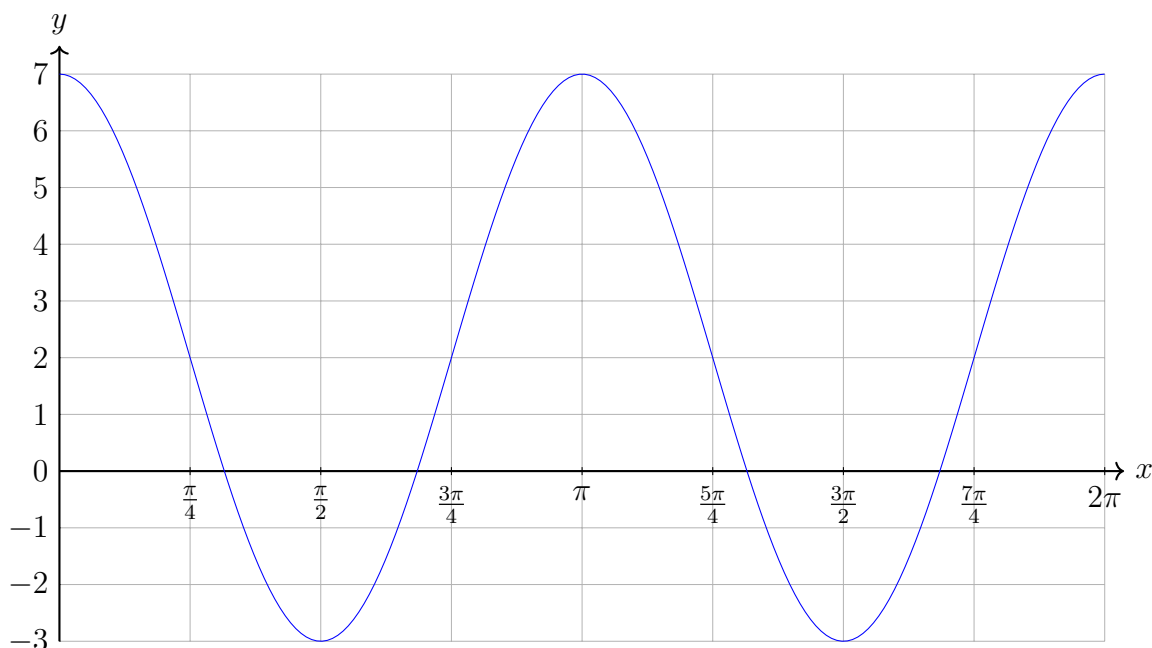
- (b) Find an equation for the graph below in the form  $y = A \sin(Bx) + D$  or  $y = A \cos(Bx) + D$  (where  $A$ ,  $B$  and  $D$  are constants).



- (c) Find an equation for the graph below in the form  $y = A \sin(Bx) + D$  or  $y = A \cos(Bx) + D$  (where  $A$ ,  $B$  and  $D$  are constants).



- (d) Find several (at least 3) trigonometric equations of the form  $y = A \sin[B(x + C)] + D$  or  $y = A \cos[B(x + C)] + D$  (where  $A$ ,  $B$ ,  $C$  and  $D$  are constants) for the graph below.



5. Evaluate the following limits analytically.

(a)  $\lim_{x \rightarrow 3} x^2 + 1$

(b)  $\lim_{x \rightarrow 4} 2x - 8$

(c)  $\lim_{x \rightarrow 2} \frac{2x + 1}{x}$

(d)  $\lim_{x \rightarrow -1} \frac{3x^2 + 1}{x - 1}$

(e)  $\lim_{x \rightarrow 5} \frac{x^2 - 5x}{x - 5}$

(f)  $\lim_{x \rightarrow -3} \frac{x^2 + 5x + 6}{x + 3}$

(g)  $\lim_{x \rightarrow 0} \frac{7x - x^2}{x}$

(h)  $\lim_{x \rightarrow -3} \frac{5x + 15}{x^2 - 9}$

(i)  $\lim_{x \rightarrow 4} \frac{2x^3 - 8x^2}{x - 4}$

(j)  $\lim_{x \rightarrow -1} \frac{x^2 + 6x + 5}{x^3 + x^2}$

(k)  $\lim_{x \rightarrow 2} \frac{x^2 + 4x - 12}{x^2 - 3x + 2}$

6. (Optional) Determine whether the following functions are continuous or discontinuous.

(a)

$$f(x) = \frac{x^2 - 25}{x - 5}$$

(b)

$$f(x) = \begin{cases} 4 - x & x \leq -1 \\ 3x + 1 & x > -1 \end{cases}$$

(c)

$$f(x) = \begin{cases} 4 - x^2 & x < -2 \\ x + 2 & x \geq -2 \end{cases}$$